

Mainul Hossain

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SUMMARY

Versatile Site Reliability Engineer with a strong background in cloud infrastructure, automation, and security remediation across large multi-subscription Azure/AWS tenants. Hands-on expertise in Kubernetes/AKS lifecycle management, Terraform, Azure Policy & Defender for Cloud remediation, and scripting (Python, Bash, PowerShell, Ruby) for automation, monitoring, and API integrations. Builds AI-assisted reliability tooling (LLM + ML forecasting/anomaly detection) with real-time dashboards and end-to-end observability; research interests include ML, deep learning, and explainable AI (XAI). Adept at cross-team collaboration and driving operational excellence across cloud and hybrid infrastructure.

WORK EXPERIENCE

Site Reliability Engineer - [Optimizely Inc.](#)
Part of **Reliability/Platform Engineering** Team

Jun 2025 - Present

- Built an AI-powered infrastructure monitoring tool and Agent Prompt Optimization workflow forecasting Black Friday readiness, with automated PowerShell performance data collection cutting manual analysis time 30%.
- Built a Terraform-managed Datadog Workflow that auto-restarts the Datadog agent on Elasticsearch via Azure RunCommand (VM/VMSS-aware, audited, on-call escalation), supporting a 99.99% SLA and cutting manual intervention 50%.
- Led a tenant-wide Defender for Cloud/Advisor security remediation program: enabled EncryptionAtHost on Linux Elasticsearch VMs across 5 global regions via Octopus Deploy PowerShell runbooks (closing a residual Guest Configuration gap via Azure Policy), and rolled out automatic periodic update assessment across 200+ VMs, resolving PowerShell 5.1 and Terraform validation blockers along the way.
- Led a full AKS cluster upgrade (v1.32 to v1.33.x) across 48 clusters ahead of Azure's platform retirement deadline, resolving stuck node-upgrade failures via manual draining; scoped the follow-on Azure Policy add-on rollout (auto-provisioning vs. Terraform) for the same fleet.
- Audited 1,430 subscriptions via Azure Resource Graph to identify SSTP VPN Gateways ahead of retirement, confirming 7 gateways for migration; migrated Azure Function Apps from Linux Consumption to Flex Consumption ahead of the Sep 2028 deadline for faster scaling and built-in network security.
- Refactored container registry authentication for two production services from hard-coded secrets to managed identity, removing a recurring downtime cause; researched a Datadog-native monitor for Azure AD service-principal credential expiration to replace a legacy Automation runbook.
- Automated Datadog health-report generation per DXP service with Cloudflare Zone integration; decommissioned 40+ unused Azure Function subnets and legacy Track Function Apps, and implemented Terraform-managed storage lifecycle policies for Event Hub data.
- Supported on-call/support rotations resolving P1 incidents (Application Gateway, API gateways, Elasticsearch cluster health) and capacity planning reviews; implemented Azure automation for .NET Core proactive monitoring, cutting investigation time 70%.
- Collaborated across SRE, Operations, and Engineering to define SLOs/SLIs and expand IaC and secure deployment practices across AWS and Azure.

Managed Services Engineer - [Optimizely Inc.](#)
Part of **Strategic Support Solutions** Team

Mar 2024 - Jun 2025

- Deployed containerized Python automation using Docker/Kubernetes for auditing and cost-optimization processes.
- Implemented infrastructure-as-code practices for Azure service optimization, reducing operational costs by 25%.
- Developed Python and Bash automation for API integrations, data extraction, and troubleshooting workflows.

- Enhanced CI/CD pipelines using GitHub Actions for microservices deployment, reducing deployment time by 30%.
- Built monitoring and alerting solutions using Azure KQL and Datadog for distributed system observability.
- Administered Linux-based systems and managed AWS/Cloudflare CDN services for high-availability architectures.
- Maintained incident management processes and provided L2/L3 support for production system outages.

Associate Managed Services Engineer - Optimizely Inc.
Part of Incident Management Team

Sep 2022 - Mar 2024

- Led incident response during high-volume traffic surges and major system outages, ensuring rapid detection and resolution.
- Managed production deployments and database changes with zero-downtime deployment strategies.
- Administered F5 load balancers and network security configurations for distributed systems.
- Served as Azure Administrator for DXP platform, maintaining 99.5% uptime across multiple regions.
- Implemented standardized incident management procedures, improving escalation processes and documentation.

Information Technology Officer - University Of Scholars
Part of Information Technology Team

Mar 2022 - Sep 2022

- Designed and maintained Linux/Windows server infrastructure for university database and web systems.
- Deployed Windows Server domain controllers and implemented organization-wide monitoring solutions.
- Managed LAN/WAN network architecture and IP device configurations for campus connectivity.

EDUCATION

May 2015 - Jan 2022 B.Sc in Computer Science - BRAC University

(GPA: 3.14/4.0)

TECHNICAL SKILLS

Programming Languages -	Python, Ruby, JavaScript, Java, C#, PHP
Cloud Platforms -	Microsoft Azure (multi-subscription tenant administration), AWS, Google Cloud Platform
Infrastructure-as-Code -	Terraform, Ansible, Azure DevOps
Scripting & Automation -	Python, Bash, PowerShell
Containerization & Orchestration -	Docker, Kubernetes, AKS cluster lifecycle management (upgrades, node draining, policy add-ons)
Cloud Security & Governance -	Azure Policy, Defender for Cloud & Advisor remediation, Guest Configuration, Managed Identity, Azure Resource Graph
AI/ML & LLM -	Prompt Engineering, LLM integration, Agent tool design, Ollama, scikit-learn, LightGBM, Prophet
Observability Tools -	Datadog, Prometheus, Grafana, Loki, Promtail, Azure KQL, ELK Stack, New Relic, Pingdom
CI/CD Platforms -	GitHub Actions, Azure DevOps, Octopus Deploy
Operating Systems -	Linux (Ubuntu, CentOS, RHEL), Windows Server
Databases -	PostgreSQL, MySQL, Azure SQL Database
Storage/Cloud Services -	Amazon S3, Azure Blob Storage
Network & Security -	F5 Load Balancers, Cloudflare CDN, VPN Gateway migration (SSTP to IKEv2/OpenVPN), Network Configuration

KEY PROJECTS

Infrastructure Self-Healing Agent (AI/ML + LLM)

[GitHub](#)

- Built a production-oriented auto-remediation platform for containerized services featuring a real-time React dashboard (Flask-SocketIO/WebSockets) and a local LLM workflow (Ollama; Llama 3.2) for RCA-style summaries, suggested remediation, and chat-based operator assistance.
- Implemented optional ML for ops signals – anomaly detection (Isolation Forest), forecasting (Prophet), failure-risk prediction (LightGBM) with PostgreSQL-backed artifacts – and packaged the platform for Docker Compose/Kubernetes (Helm/Kustomize) with Prometheus, Grafana, and Loki observability.

.NET Core Monitoring & Diagnostics (Azure App Service)

[GitHub](#)

- Built a master Bash runbook to monitor thread count, response time, and outbound connections for .NET Core workloads on Linux, automating dump/trace capture (dotnet-dump, dotnet-trace, dotnet-counters) with duplicate-prevention locking and Azure Blob upload (AzCopy) with retries and instance-aware log rotation.

Cloudflare DNS Records Export Tool (Parallel Extraction)

[GitHub](#)

- Built a Python exporter to inventory DNS records across Cloudflare zones (filtered by account, with pagination and retry logic), using ThreadPoolExecutor parallelism and Docker-friendly output defaults for CI/CD and container runs.

Cloudflare WAF Rules Extractor (Security Audit Automation)

[GitHub](#)

- Automated extraction of Cloudflare custom firewall rules across all zones with action-based filtering (e.g., skip/block/challenge), producing audit-ready CSV reports via parallel zone processing and resilient API retry/backoff behavior.

Pingdom API Automation & Monitoring (Async)

[GitHub](#)

- Built an asyncio/aiohttp tool for concurrent hostname checks using the Pingdom API with structured CSV reporting for ops workflows.

CERTIFICATIONS & AWARDS

Microsoft Certified: Azure Administrator Associate
Microsoft Corporation - **Credential ID MS0995597094**

Issued Jun 2024

Spot Award
Optimizely Spot Award for "Never Stop Improving"

Issued Aug 2023

Spot Award
Optimizely Spot Award for "Getting it done together"

Issued Apr 2023

Introduction to Programming Using Java
LICT Project, ICT Division, Government of Bangladesh - **Credential ID G047514**

Issued Mar 2018